Title and author(s).

The title should reflect your specific research questions (not "CSE 163 Project"). Your name(s) should be clearly displayed near the top of the document as well.

NHL Draft Picks: Rowen Bertelsen, Srinidhi Muralidharan, Lily Yao

Research Questions.

Give a numbered list of 3 or more research questions. Each research question should be 1 to 3 sentences long, allow for a clear and definitive answer, and provide significantly more specific details to your project for your TA reviewer.

- Does height and weight matter the most for each position (goalie, defense, forward)?
- Which parts of the world are most represented in the NHL? (Heatmap)
- What factors (league, height, weight) play the biggest role in success for players drafted in the NHL?
- Which junior leagues are drafted into the NHL the most and which ones do the best in the NHL?
- Which NHL teams have had the most success in drafting based on plus and minus? Or point shares?
- Based on the first pick every year from 1998 to 2008, how well (point shares) did the first pick player comparatively do?

Motivation

Expand on your research questions by providing context about why you care about the problem. How does knowing the answers affect the world or our understanding of it?

Like many sports leagues, the NHL holds a draft each year in which each of the 32 (previously fewer) NHL teams picks players round by round that they will have the rights to if the player decides to enter the NHL. Sports is a growing industry, and a lot of a team's success is determined by their ability to successfully scout players with the highest potential. In our project, we will examine how we can use statistics to draft players that we predict will have a higher success rate in the NHL. Knowing which factors contribute most to a draft pick's future success can help the NHL grow as a league and help specific teams win.

Data Setting

Describe the dataset that you will use and **include a link to obtain the data**. If a datasheet is available, describe 3 ways the context of the dataset might complicate or deepen your analysis. If a datasheet is not available, explore the dataset to identify at least 3 example data entries that raise questions about the data setting.

https://www.kaggle.com/datasets/mattop/nhl-draft-hockey-player-data-1963-2022
This dataset is collected from Sports Reference. It contains every player drafted in the NHL draft from 1963 to 2022. We found columns such as order of player was drafted, amateur team drafted from, and point shares really important to our analysis.

https://www.nhl.com/stats/skaters
This data is collected from the NHL and provides game statistics of all players who have played in the NHL.

https://github.com/liuyejia/Model_Trees_Full_Dataset/blob/master/elite_prospects_skaters_stats_1998_2008_original.csv
This data set was collected by Elite Prospects and contains game data for the year before the player was drafted for all the players drafted into the NHL from 1998 to 2008.

Challenge Goals

Select 2 challenge goals that you are planning to meet from the list below. Justify why you think your project will meet each goal. Bold each challenge goal name to make it clear which one you are talking about.

We would like to use a **new library** such as Altair or Plotly to create a heat map and use **multiple datasets** to combine into one csv file to analyze. In particular, our heat map will be a map of the world and will be used to specify players that're from countries like the US, Canada, Russia, Sweden, and much more. We would like to combine columns like "pointshare" and overall draft pick, as well as all the data from the years 1998 to 2008 from our Kaggle data. With NHL data, columns like "points per game", specific games, and player statistics will be used. From the github dataset, we will use information about players' statistics in junior leagues.

Method

Outline your analysis as a step-by-step process in sufficient detail for someone else to independently reproduce your analysis without asking you any additional questions about your process. Think of this as writing documentation for your project.

- Break down each step into one or more functions per step. Identify each function as a data-manipulation function or a plotting function.
- Explain your plan to test each function. Test each data-manipulation function on small, manually-verifiable testing datasets. Plotting functions do not need to be tested.
- Indicate how the result would lead you to a certain conclusion about one or more research questions. Highlight connections to challenge goals.

- **Clean the three data and merge them together.**
  Drop the unwanted columns. Drop the player rows we aren't interested in. Create a new dataset from the three data sets.
- **Answering different questions and Create Visualization:**
  We would like to use Pandas and Seaborn to plot facegrids.
    - Does height and weight matter the most for each position (goalie, defense, forward)?
        - Groupby the positions and compare the heights and weights.
    - Which parts of the world are most represented in the NHL? (Heatmap)
        - Groupby nationality and compare counts
    - What features (league, height, weight, point_shared) play the biggest role in success for players drafted in the NHL?
        - Groupby overall_pick (order of being picked) and compare the heights, weights, and leagues
        - Do a **linear regression/random forest regression** and compare the coefficients of each feature. See which one weights more.
    - Which junior leagues are drafted into the NHL the most and which ones do the best in the NHL?
        - Groupby the junior leagues and compare the overall_pick (the order of being picked)
    - Which NHL teams have had the most success in drafting based on plus and minus? Or point shares?
        - Groupby league teams and compare point shares
    - Based on the first pick every year from 1998 to 2008, how well (point shares) did the first pick player comparatively do?
        - Drop the players from the data other than the first-pick player, then compare the point shares

Plan

Describe your plan to divide the project into at least 3 (but no more than 10) independent, high-level tasks, each with an estimate of the time in hours required for each task. Then, describe your workflow for developing code, testing code, and coordinating work—particularly how team members can support each other in case one task is unexpectedly challenging. All team members must contribute equally to the deliverables.

1. Clean and merge the three dataset correctly for future analysis: **5 hours**
2. Creating visualizations for analysis between different features and outcomes: **10 hours**
3. Creating linear regression models/random forest models to predict player-drafted: **15 hours**
4. Create tests for each function. **5 hours**

We are hoping to split the work later because we'll work on the project together in the initial stages and then partition the coding assignments. However, when we partition the code, we will split it evenly and make sure to help each other while giving each other space to problem solve and work alone.